STRUCTURE & CLASSIFICATION OF AMINO ACIDS

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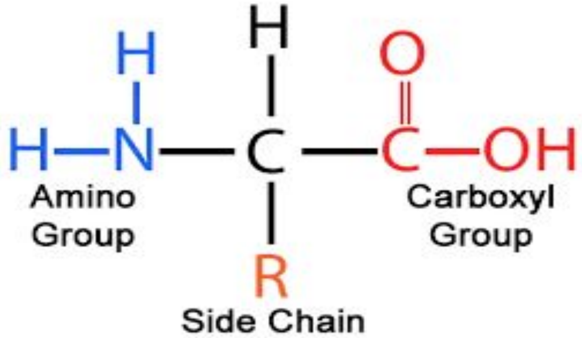

Learning Objectives

- Describe diagrammatically the general structure of an Amino Acid with example.
- Give a general classification of Amino Acids with examples and biomedical significance of each type.
- Describe all Essential and Non essential Amino Acids with their types, biomedical and clinical significance.
- What is meant by Glucogenic and ketogenic Amino Acid? Differentiate them and give examples of both types with their biomedical significance.

INTRODUCTION

- Proteins are the most abundant organic molecules of the living system.
- They occur in the every part of the cell and constitute about 50% of the cellular dry weight.
- Proteins form the fundamental basis of structure and function of life.
- In 1839 Dutch chemist G.J.Mulder while investing the substances such as those found in milk, egg, found that they could be coagulated on heating and were nitrogenous compounds.

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- These molecules are crucial for structure, regulation, and enzymatic functions, making them the dominant macromolecule in cellular dry mass.

- They are *nitrogenous "macromolecules "composed of many amino acids*.

- The term is derived from Greek word *Proteios means "primary", or "holding first place" or* "pre-eminent".

Biomedical importance of proteins:

- ❑ Proteins are the main structural components of the cytoskeleton.
- ❑ Bio chemical catalysts known as enzymes are proteins.
- ❑ Proteins known as immunoglobulins serve as the first line of defence against bacterial and viral infections.

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- Several hormones are protein in nature.
- Structural proteins like actin and myosin are contractile proteins and help in the movement of muscle fibre.
- The transport proteins carry out the function of transporting specific substances either across the membrane or in the body fluids.

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- Storage proteins bind with specific substances and store them, e.g. iron is stored as ferritin.
- Few proteins are constituents of respiratory pigments and occur in electron transport chain,
  - e.g. Cytochromes, hemoglobin, myoglobin
- Under certain conditions proteins can be catabolized to supply energy.
- Proteins by means of exerting osmotic pressure help in maintenance of electrolyte and water balance in the body.

AMINO ACIDS

Definition:

"The building blocks of proteins are known as Amino Acids"

OR

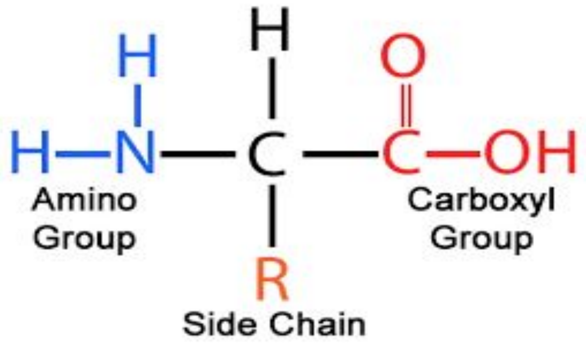
"A simple organic compound containing both a carboxyl (—COOH) and an amino (—NH_2) group is called Amino Acid"

AMINO ACIDS

- Amino acids are a group of organic compounds containing two functional groups – *amino and carboxyl.*
- The amino group ($-NH_2$) is basic while the carboxyl group (-COOH) is acidic in nature.
- There are about 300 amino acids occur in nature.
- Only 20 of them occur in proteins.

STRUCTURE OF AMINO ACIDS

- Each amino acid has 4 different groups attached to α- carbon (which is C-atom next to COOH).
- These 4 groups are : amino group, COOH gp, Hydrogen atom and side Chain (R)



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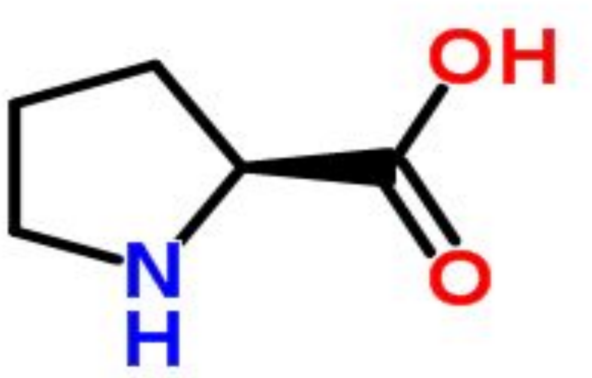
Exception to this rule is **Proline** which contains ***Imino*** group (NH) instead of ***Amino*** group



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- Amino acids differ in their properties due to differing side chains, called R groups.
  - In proteins , almost all of the these carboxyl and amino groups are combined in peptide linkage and are not available for chemical reaction except for hydrogen bond formation .
  - Thus it is the nature of the side chains that ultimately dictates the role , an amino acid plays in a protein.

- There are Standard and Non Standard Amino Acids.

STANDARD AMINO ACIDS (PROTEINOGENIC)

- They take part in protein synthesis
- More then 300 AA have been described
- Only 20 AA are used to build proteins.
- These 20 AA are called primary or standard AA
- There are essential and non essential AA

Phenylalanine	Histidine
Tryptophan	Arginine
Valine	Lysine
Threonine	Leucine
Leucine	Alanine
Isoleucine	Cysteine
Methionine	Glutamic Acid
Serine	Aspartic Acid

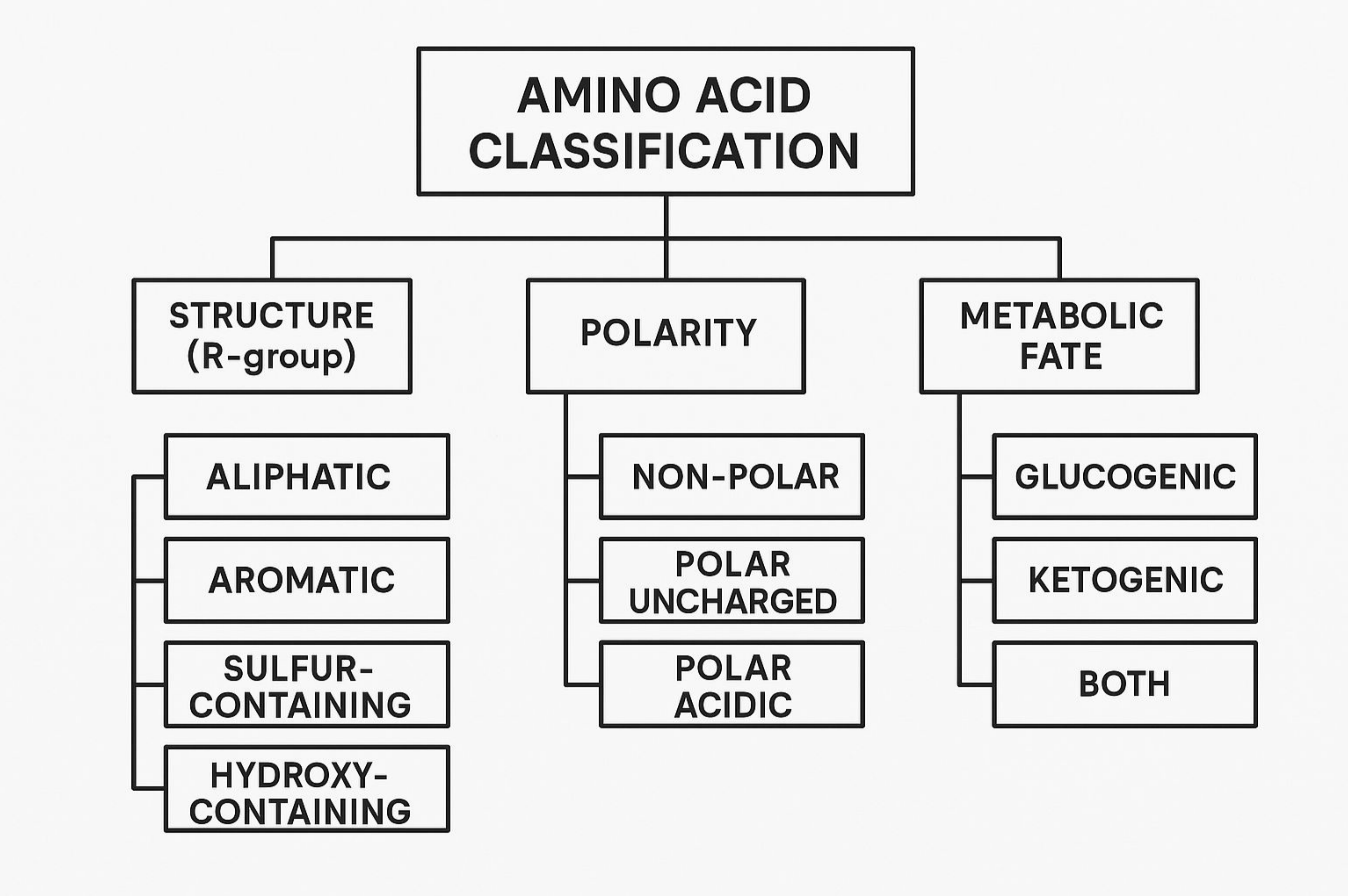
Non Standard Amino Acids

These Amino acids do not take part in the protein synthesis but play important role in the body.

- **Citrulline**
- **Ornithine**
- **Taurine**
- **DOPA**
- **GABA**

CLASSIFICATION OF AMINO ACIDS

- Nutritional Classification
- Classification by R group
- Metabolic Classification



NUTRITIONAL CLASSIFICATION

1- Essential Amino Acids.

2- Semi-essential: formed in the body but not in sufficient amount for body requirements especially in children. Arginine and histidine are semi-essential .

3- Non-essential can be synthesized in the body.

Essential or indispensable amino acids

- The amino acids which cannot be synthesized by the body and need to be supplied through the diet are called essential amino acids.

- They are required for proper growth and maintenance of the individual.

- Their deficiency affects growth, health and protein synthesis.

The 10 essential amino acids are: (Mnemonic PVT TIM HaLL)

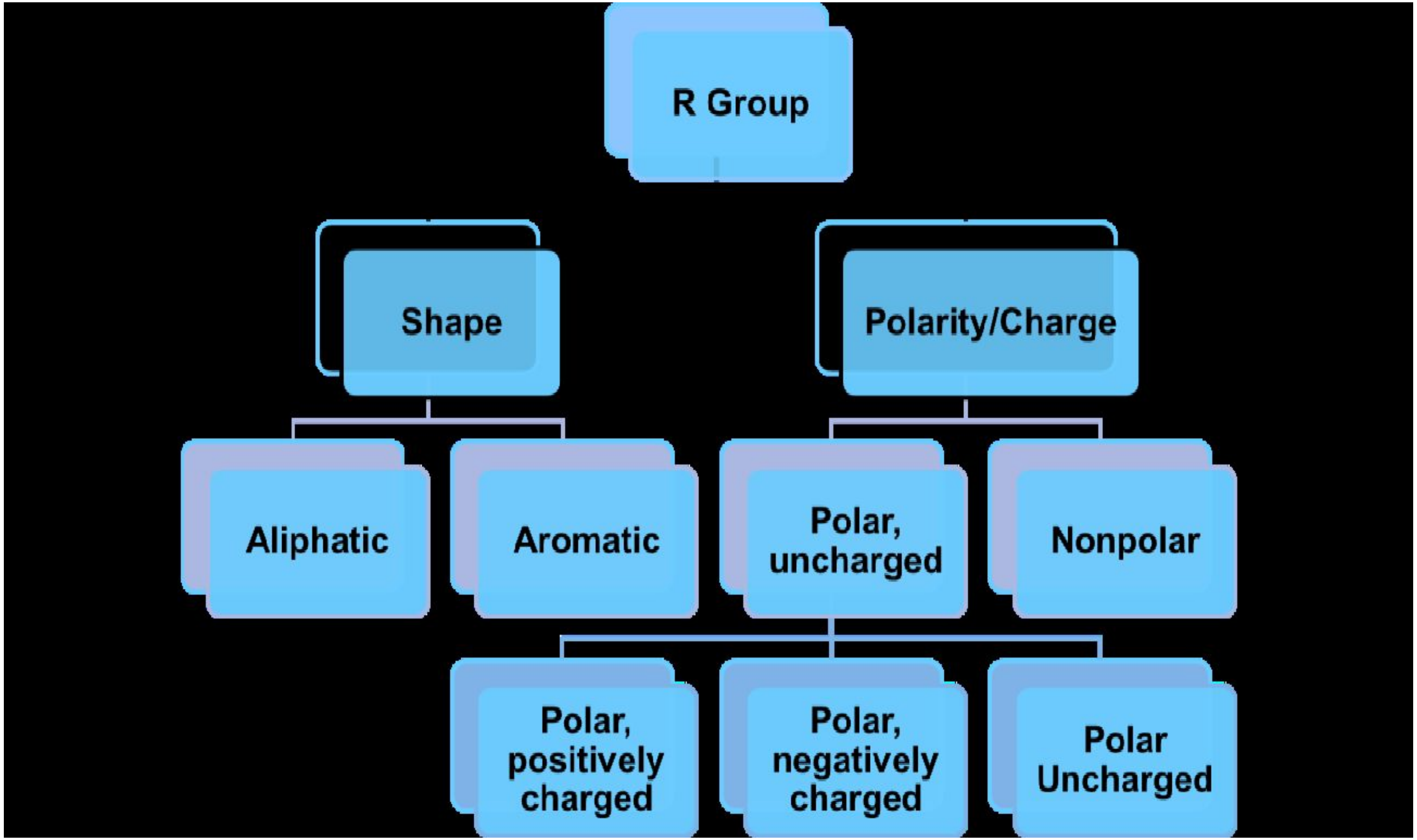
- 1. Arginine -Semi essential
- 2. Valine
- 3. Histidine -Semi essential
- 4. Isoleucine
- 5. Leucine
- 6. Lysine
- 7. Methionine
- 8. Phenyl alanine
- 9. Threonine
- 10. Tryptophan

NON – ESSENTIAL AMINO ACIDS:

- The body can synthesize about 10 amino acids to meet the biological needs, hence they need not be consumed in the diet.

- These are Glycine, Glutamate, Glutamine, Alanine, Aspartate, Aspargine, Serine, Proline Cysteine and Tyrosine (GGGAAAS [gas] P CT).

Classification by R group



CLASSIFICATION BY R GROUP

- Amino acid classification based on the structure
- The 20 standard amino acids found in protein structure are divided into seven distinct groups.

- 1. Aliphatic amino acids:
- 2. Hydroxyl group containing amino acids:
- 3. Sulfur containing amino acids:

- 4. Aromatic amino acids
- 5. Acidic amino acids and their amides
- 6. Basic amino acids
- 7. Imino acids

Classification of Amino Acids

Amino Acids with Aliphatic Side Chain

Glycine

Alanine

Valine

Leucine

Isoleucine

Amino Acids with Side Chain Having OH Group

Serine

Threonine

Tyrosine

Classification of Amino Acids

Amino Acids with Side Chain Containing Sulphur Atom

Cysteine

Cystine

Methionine

Amino Acids with Aromatic Side Chain

Phenylalanine

Tyrosine

Tryptophan

Classification of Amino Acids

Amino acids with Acidic Side Chains

These are mono amino dicarboxylic

Aspartic acid

Glutamic acid

Amino acids with Basic Side Chains

These are diamino monocarboxylic

Arginine

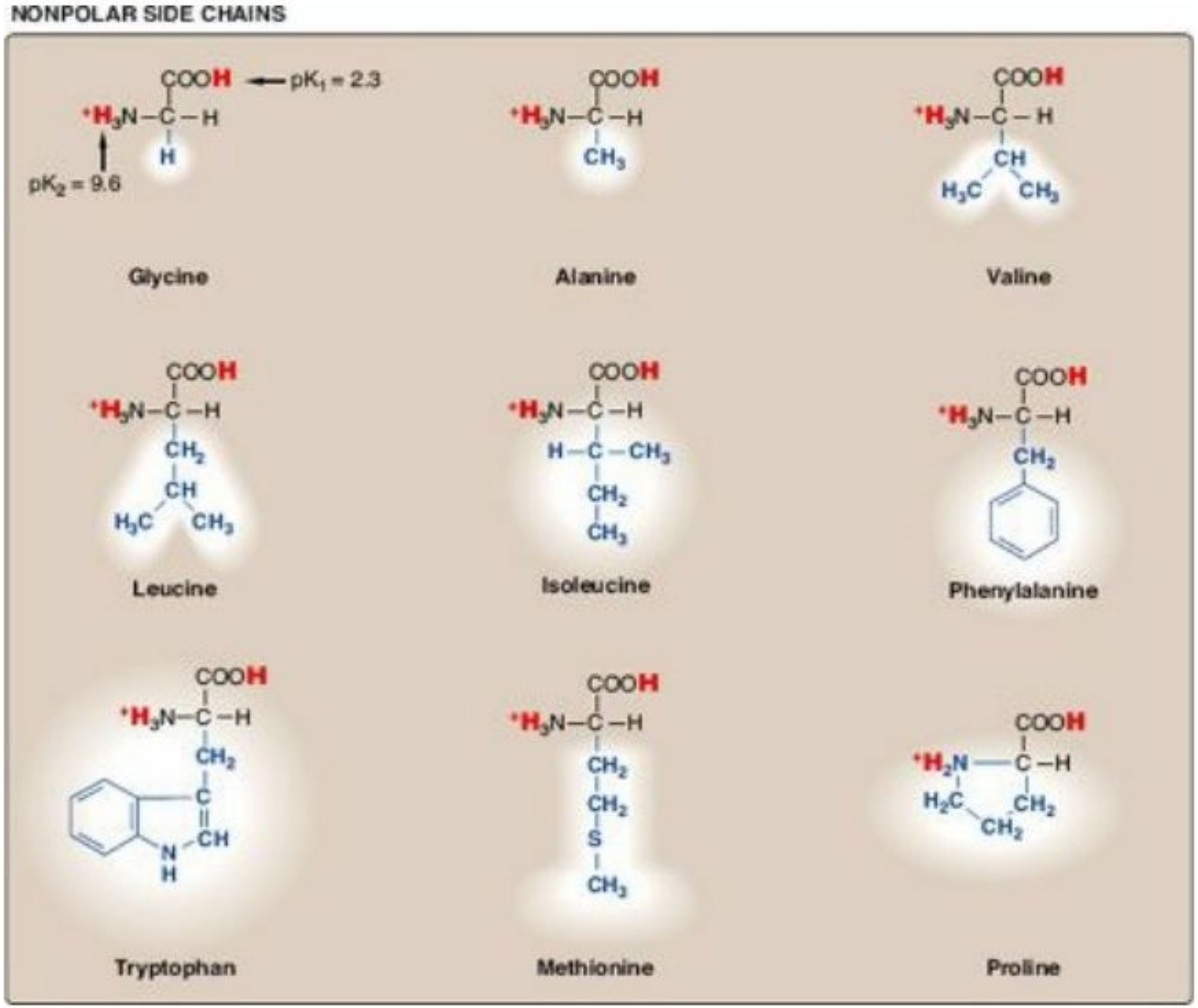
Histidine

Lysine

CLASSIFICATION OF AMINO ACIDS BASED ON POLARITY:

- Amino acids are classified into 4 groups based on their polarity.

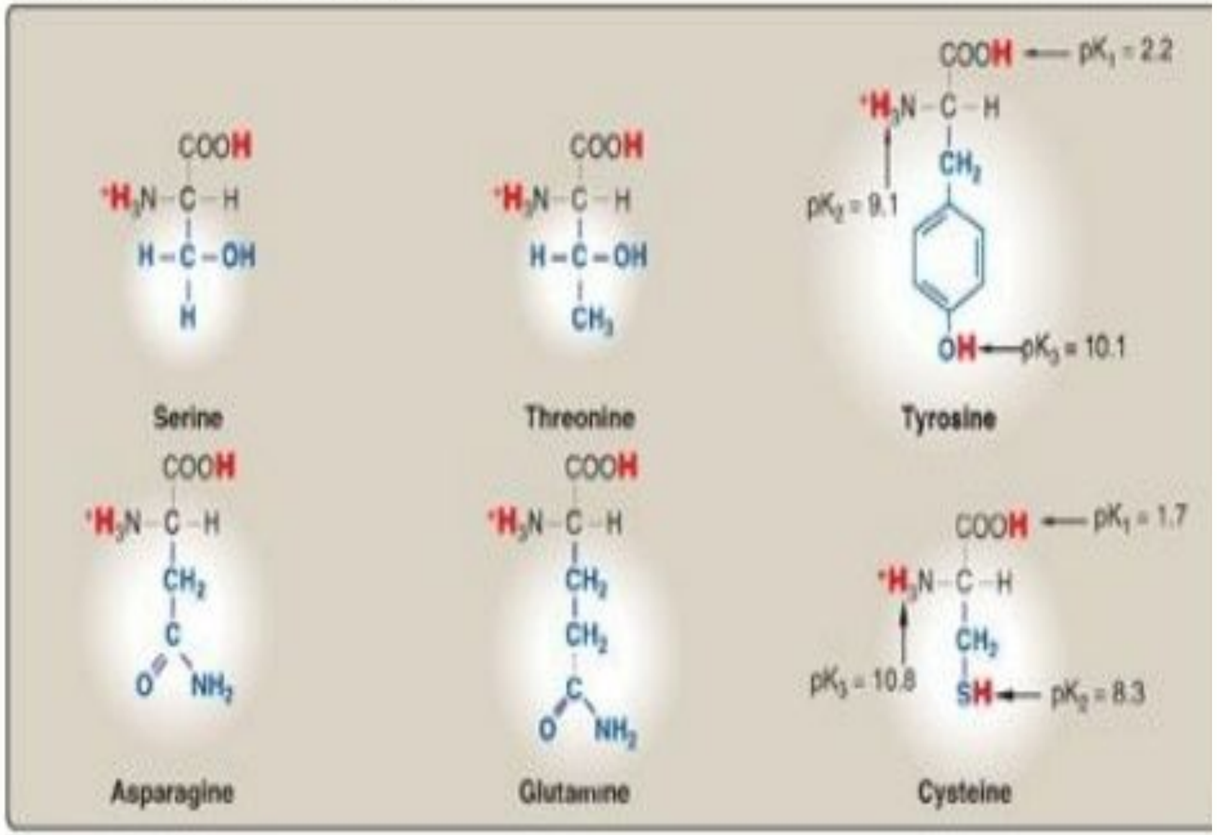
1. Non – polar amino acids:
   - ❑ These amino acids are also referred to as hydrophobic.
   - ❑ They have no change on R-group.


NONPOLAR SIDE CHAINS
pK1 = 2.3
pK2 = 9.6
Glycine
Alanine
Valine
Leucine
Isoleucine
Phenylalanine
Tryptophan
Methionine
Proline

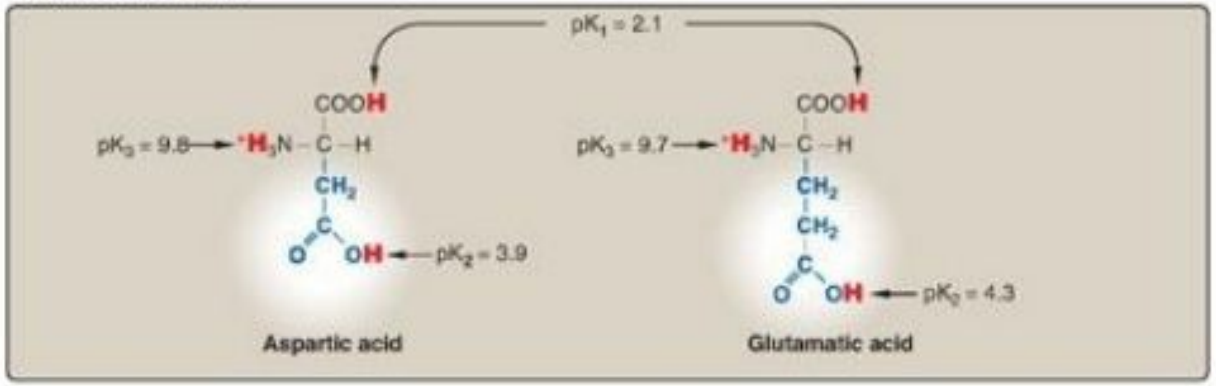
2. Polar amino acids with no charge on R-group:

- These amino acids, as such, carry no charge on the R-group.
- They however possess groups such as hydroxyl, sulfhydryl and amide and participate in hydrogen bonding of protein structure.

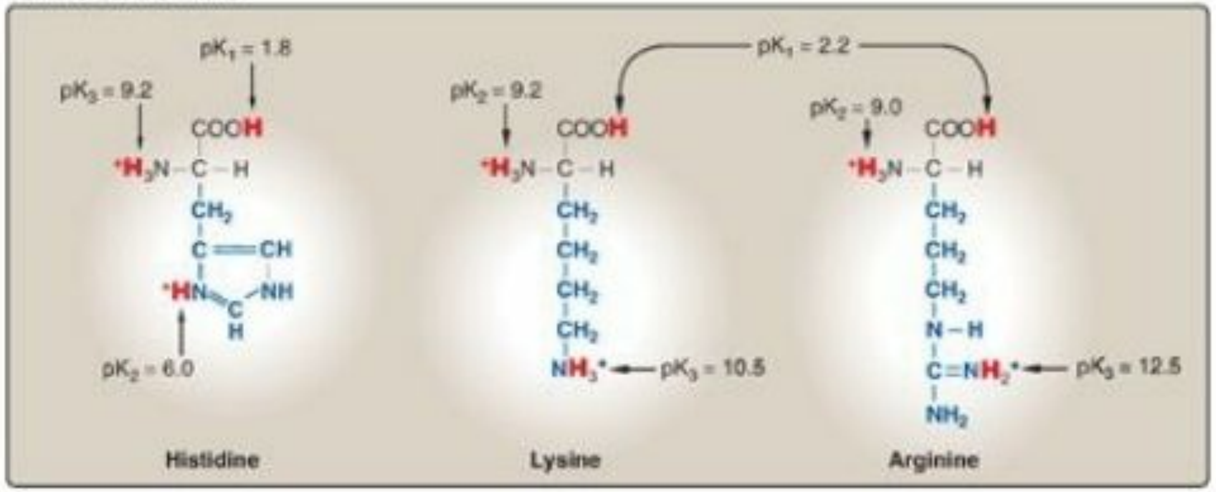
UNCHARGED POLAR SIDE CHAINS



ACIDIC SIDE CHAINS



BASIC SIDE CHAINS



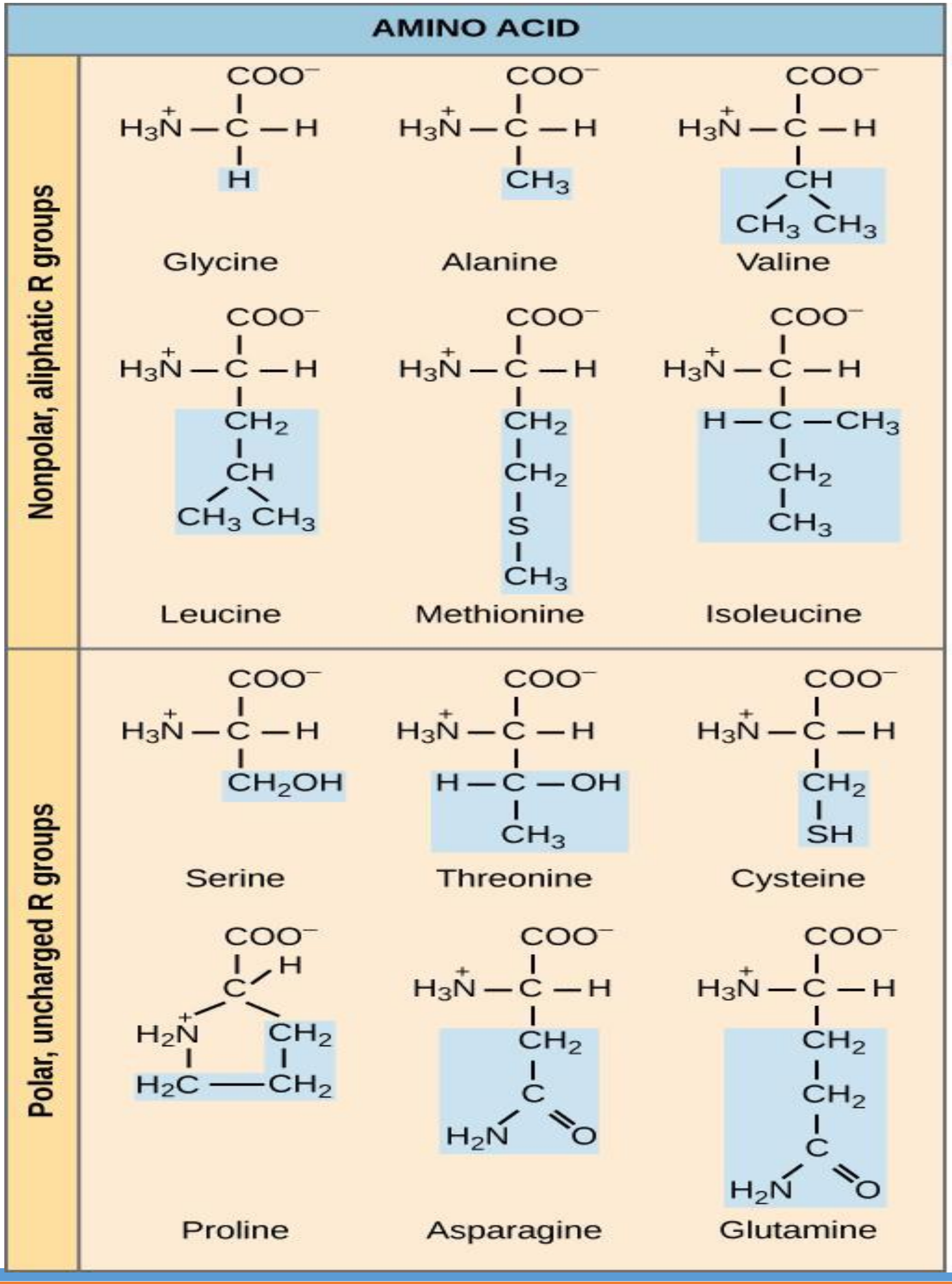
3.Polar amino acids with positive R- group:

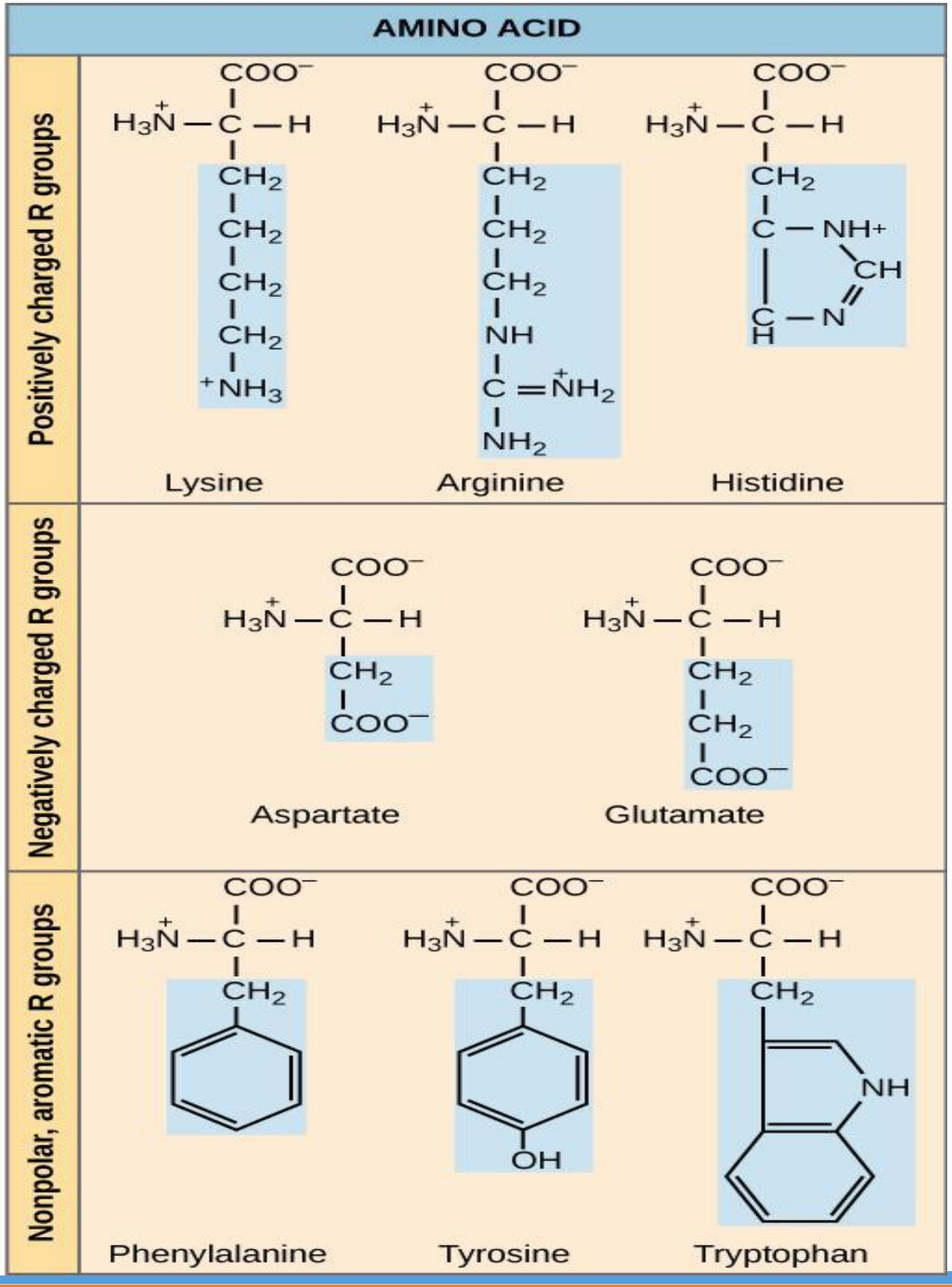
The three amino acids:

- Lysine
- Arginine
- Histidine

4. Polar amino acids with negative R- group:

- The dicarboxylic mono amino acids – aspartic acid and glutamic acid are considered in this group.


AMINO ACID
Nonpolar, aliphatic R groups
Glycine
Alanine
Valine
Leucine
Methionine
Isoleucine
Polar, uncharged R groups
Serine
Threonine
Cysteine
Proline
Asparagine
Glutamine


AMINO ACID
Positively charged R groups
Lysine
Arginine
Histidine
Negatively charged R groups
Aspartate
Glutamate
Nonpolar, aromatic R groups
Phenylalanine
Tyrosine
Tryptophan

Based on their metabolic fate:

- The carbon skeleton of amino acids can serve as a precursor for the synthesis of glucose (glucogenic) or ketone bodies (ketogenic) or both.

- From metabolic veiw point, amino acids are divided into three groups

Glycogenic amino acids:

- These amino acids can serve as precursor for the formation of glucose or glycogen. E.g.,alanine, aspartate, glycine, methionine etc.

Ketogenic amino acids:

- Ketone bodies can be synthesized from these amino acids. Two amino acids leucine and lysine are ketogenic.

Glycogenic and ketogenic Amino acids

The four amino acids are precursors for the synthesis of glucose as well as ketone bodies.

- ❑ Phenyl alanine
- ❑ Isoleucine
- ❑ Tryptophan
- ❑ Tyrosine

Glucogenic and Ketogenic Amino Acids

Glucogenic	Both Glucogenic and Ketogenic	Ketogenic
Aspartate	Isoleucine	Leucine
Asparagine	Phenylalanine	Lysine
Alanine	Tryptophan	
Glycine	Tyrosine	
Serine		
Threonine		
Cysteine		
Glutamate		
Glutamine		
Arginine		
Proline		
Histidine		
Valine		
Methionine		

The ***twenty common amino acids*** are often referred to using three-letter abbreviations. The structures, names, and abbreviations for the twenty common amino acids are shown below. Note that they are all ***α*-amino acids**.

Each amino acid, aside from its name, has a three letter abbreviation and a one letter code.

Nomenclature

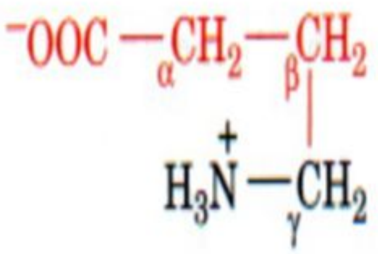
Amino Acid	3 letter code	1 letter code	Amino Acid	3 letter code	1 letter code
Glycine	Gly	G	Threonine	Thr	T
Alanine	Ala	A	Cysteine	Cys	C
Valine	Val	V	Tyrosine	Tyr	Y
Leucine	Leu	L	Asparagine	Asn	N
Isoleucine	Ile	I	Glutamine	Gln	Q
Methionine	Met	M	Aspartic Acid	Asp	D
Proline	Pro	P	Glutamic Acid	Glu	E
Phenyl alanine	Phe	F	Lysine	Lys	K
Tryptophan	Trp	W	Arginine	Arg	R
Serine	Ser	S	Histidine	His	H

Amino Acid Derivatives:

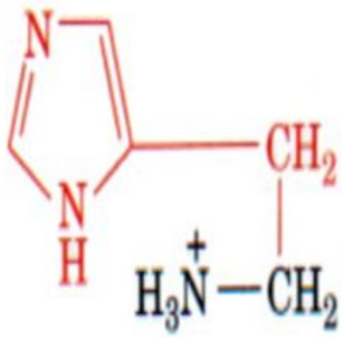
- Amino acid derivatives are bioactive compounds formed through chemical or metabolic modifications (e.g., methylation, decarboxylation) of amino acids, playing crucial roles in metabolism, neurotransmission, and physiological regulation.

- Key examples include creatine (energy), serotonin (mood), dopamine/epinephrine (signaling), taurine (antioxidant), and GABA.

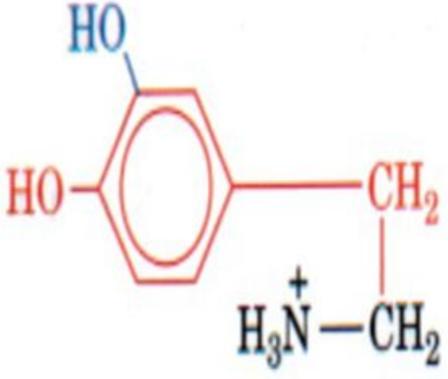
- Histamine mediates parts of the immune response.



γ-Aminobutyric acid (GABA)



Histamine



Dopamine

Functions of Amino Acids:

- Apart from being the monomeric constituents of proteins and peptides, amino acids serve variety of functions.

- Some amino acids are converted to carbohydrates and are called as **glucogenic amino acids**.

Specific amino acids give rise to specialised products, e.g.

- **Tyrosine**- produced from Phenylalanine, forms hormones such as thyroid hormones, (T3, T4), epinephrine and norepinephrine and a pigment called melanin.

- **Tryptophan** can synthesise a vitamin called niacin.

- **Glycine, arginine and methionine** synthesize creatine.

- Glycine and cysteine help in **synthesis of Bile salts**.
- Glutamate, cysteine and glycine synthesis **glutathione**.
- **Histidine** changes to **histamine** on decarboxylation.
- **Serotonin** is formed from tryptophan.

- Pyrimidines and purines use several amino acids for their synthesis such as aspartate and glutamine for pyrimidines and glycine, aspartic acid, Glutamine and serine for purine synthesis.

- Methionine transfers methyl group to various substances by transmethylation.

- Cystine and methionine are sources of Sulphur.


Thank you!